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IL MANUALE DI

GEVINO opto PNP v1.4

Compatible Arduino Zero
Manual v5

ITA
I prodotti GEVINO sono stati progettati per ambienti civili, domestici, aziendali o industriali. Hanno ingressi e uscite ad alta tensione, alta corrente, protetti contro disturbi, scariche elettrostatiche, campi magnetici, carichi induttivi, extra tensioni e correnti.

Sono conformi alle direttive CE.

E' possibile personalizzarle per una quantità superiore a 50 pezzi.

Per programmare Gevino è sufficiente collegarlo al proprio computer con un cavo USB, scrivere qualche istruzione, e premere il tasto «Upload». E' ricchissimo di librerie per un' infinità di moduli e sensori. Le istruzioni sono abbastanza semplici e comprensibili.

Il software di programmazione, chiamato anche IDE (Ambiente Integrato di Sviluppo), contiene numerosi esempi da cui prendere spunto per scrivere i propri listati.

ENG
GEVINO products have been designed for civil, domestic, corporate or industrial environments. They have high voltage and high current inputs and outputs, protected against disturbances, electrostatic discharges, magnetic fields, inductive loads, extra voltages and currents.

They comply with EC directives.

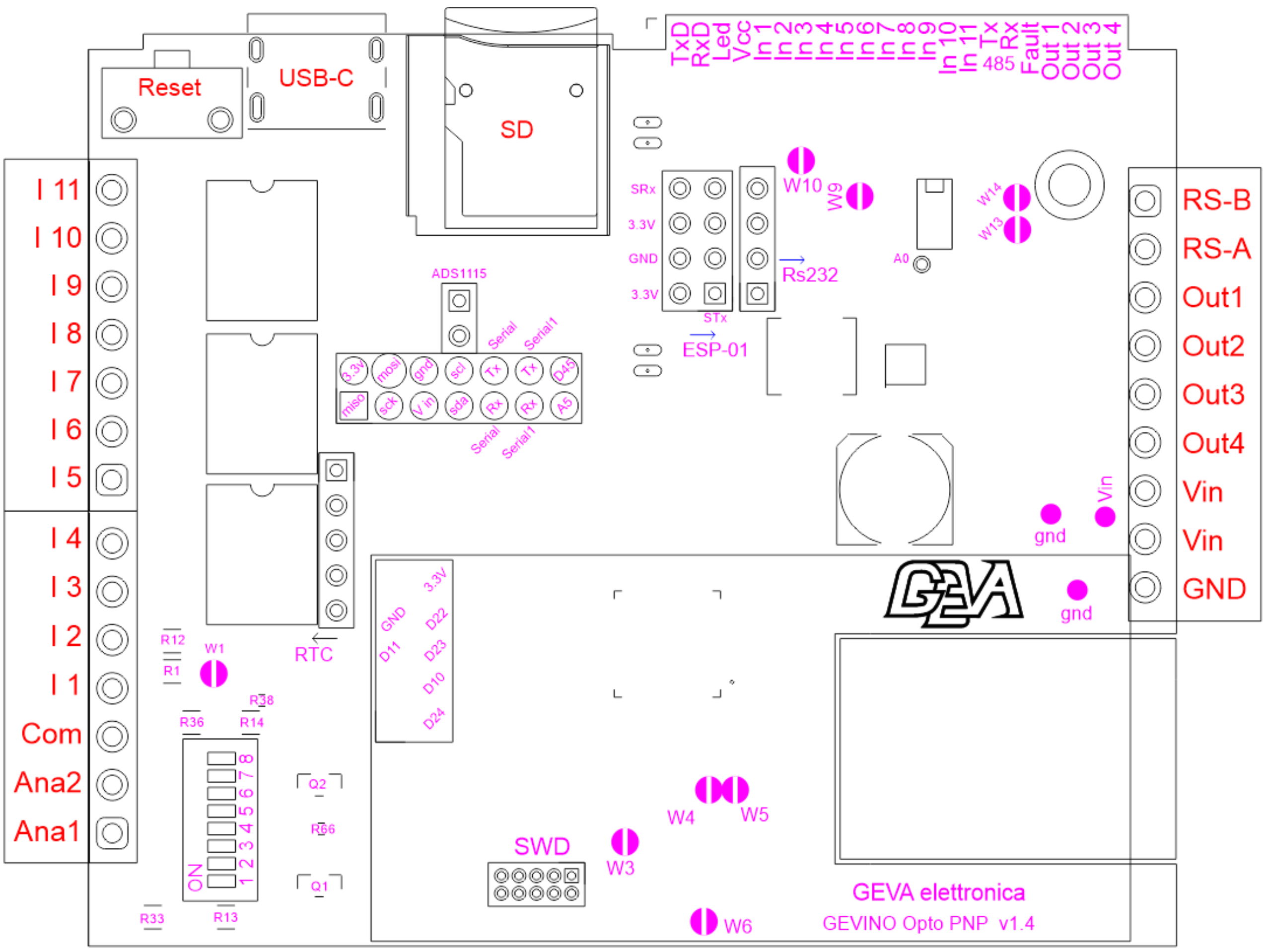
It is possible to customize them for more than 50 pieces.

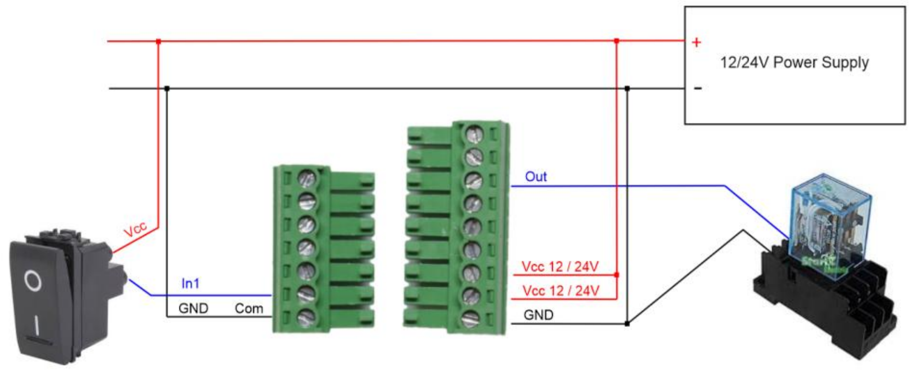
To program Gevino simply connect it to your computer with a USB cable, write some instructions, and press the «Upload» button. It is very rich in libraries for an infinity of modules and sensors. The instructions are quite simple and understandable.

The programming software, also called IDE (Integrated Development Environment), contains many examples from which to take inspiration to write their own listings.

Changelog
V1.1 cap for SIM
V1.2 NTC resistor
    P11 from ESP05 to RS232
V1.3 A0 <-> A1
V1.4 DIP Swith

Hardware





Dip Switch

Ana1 (A0)	Dip 1	Dip 2	Dip 3	Dip 4	Dip 5	
10V	off	off	off	off	on	default
1V	off	off	on	off	on	
20mA	on	off	on	off	on	
Out Mos NPN	off	on	off	off	off	
1V + 10K pullup	off	off	on	on	off	

Ana2 (A2)	Dip 6	Dip 7	Dip 8	
10V	off	off	off	default
1V	off	off	on	
20mA	on	off	on	
Out Mos NPN	off	on	off	

Bootloader

You can update the program in the GEVINO from the Internet or SD card requesting a dedicated bootloader.
Si può aggiornare il programma nel GEVINO da internet o SD card richiedendo un bootloader dedicato.

Specs / Specifiche

- Vcc from 7 to 40V
- 4 Out PNP 2.5A, short circuit protection. (Fault Red Led on front, also readable losing one input)
- 11 Optoisolated Input, High level 5 to 40V, -5 to -40V, Low level -2.5 to +2.5V, DC or AC.
- All input can be interrupt. Except In05 and In11
- 10Khz max input frequency
- CPU ATSAMD21G18 - Arduino ZERO M0 compatible.
- 48 Mhz, 32-bit ARM® Cortex®-M0+ processor.
- Flash 256KB, RAM 32K, Optional FRAM I2C 32Kbyte
- -40°C +85°C Working temperature

Utilization / Utilizzo

ARDUINO IDE https://www.arduino.cc/en/main/software
Install zero Compiler https://www.arduino.cc/en/Guide/ArduinoZero
Choose Board «Arduino/Genuino Zero (Native USB Port)»
Use the Arduino Zero reference guide.

3 enclosures, of different thicknesses:
3 contenitori, con differente spessore:

Thin 17.5mm black, for the only card without accessories.
Medium 22.5mm green,for accessories such as Ethernet, RTC, Wi-Fi, Bluetooth, GSM modem.
Thick 35mm black, for accessories such as RS232 on DB9 female, Li-Po battery with battery charger.

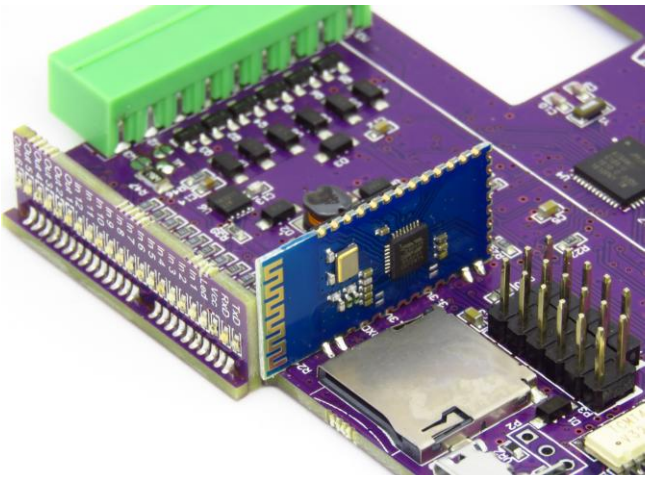
Sottile 17.5mm nera, per la sola scheda senza accessori.
Media 22.5mm verde, per accessori quali l'Ethernet, l'RTC, il Wi-Fi, il Bluetooth, il modem GSM.
Spessa 35.0mm nera, per accessori quali RS232 su DB9 femmina, Batteria Li-Po con caricabatteria.

SerCom

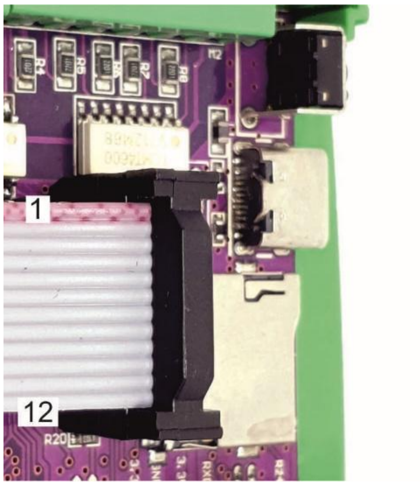
Six Serial Communication Interfaces, each configurable to operate as either:
• USART with full-duplex and single-wire half-duplex configuration
• I2C up to 3.4MHz
• SPI
• LIN slave
https://www.arduino.cc/en/Tutorial/SamdSercom
https://learn.adafruit.com/using-atsamd21-sercom-to-add-more-spi-i2c-serial-ports/overview

SerialUSB	USB	CS Addr	Reset	Int	Tx En
Serial	ESP-WiFi Bluetooth				
Serial1	RS485				A0
I2C	RTC DS3231				
“	OLED				
SPI	SD card reader	12			
“	W5500 ETH	10	11	---	
“	CAN BUS	10		11	

HC-06 -07 -08 Bluetooth on Serial:



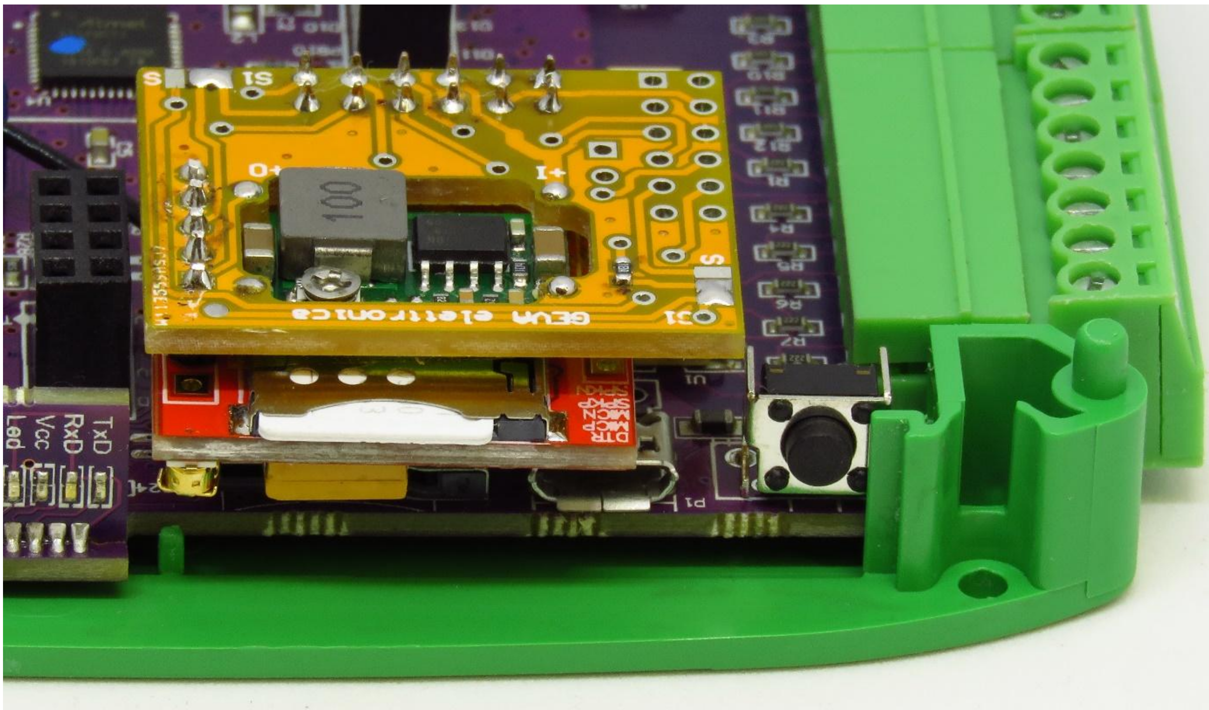
Optional Flat Cable on P3:



1- MISO – D22
2- 3.3V
3- SCK – D24
4- MOSI – D23
5- Vin (12v/24v) After Diode
6- GND
7- SDA – D20
8- SCL – D21
9- Serial RxD – D25 - ESP WiFi - Bluetooth
10- Serial TxD – D26 - ESP WiFi - Bluetooth
11- Serial1 RxD – D0- PA11 - Rs485
12- Serial1 TxD – D1 - PA10 – Rs485
13- A5 Analog, need to open W5 Jumper
14- D45-SWDIO input/output, programming pin, cannot be high at boot.

For use Serial1 RxD open W9 Jumper (Driver Rs485)
9 & 11 RxD → RxD of RS485 & RS232 module → TxD of SIM800L modem, ESP
10 & 12 TxD → TxD of RS485 & RS232 module → RxD of SIM800L modem, ESP

SIM800L modem On Serial or Serial1:



Change to Nextion library:

Line 35 of NextUpload.cpp, comment.
//#include <SoftwareSerial.h>

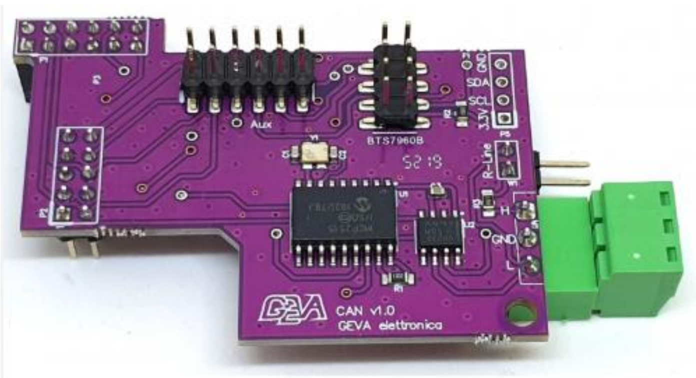
Line 29..37 of NexConfig.h, change

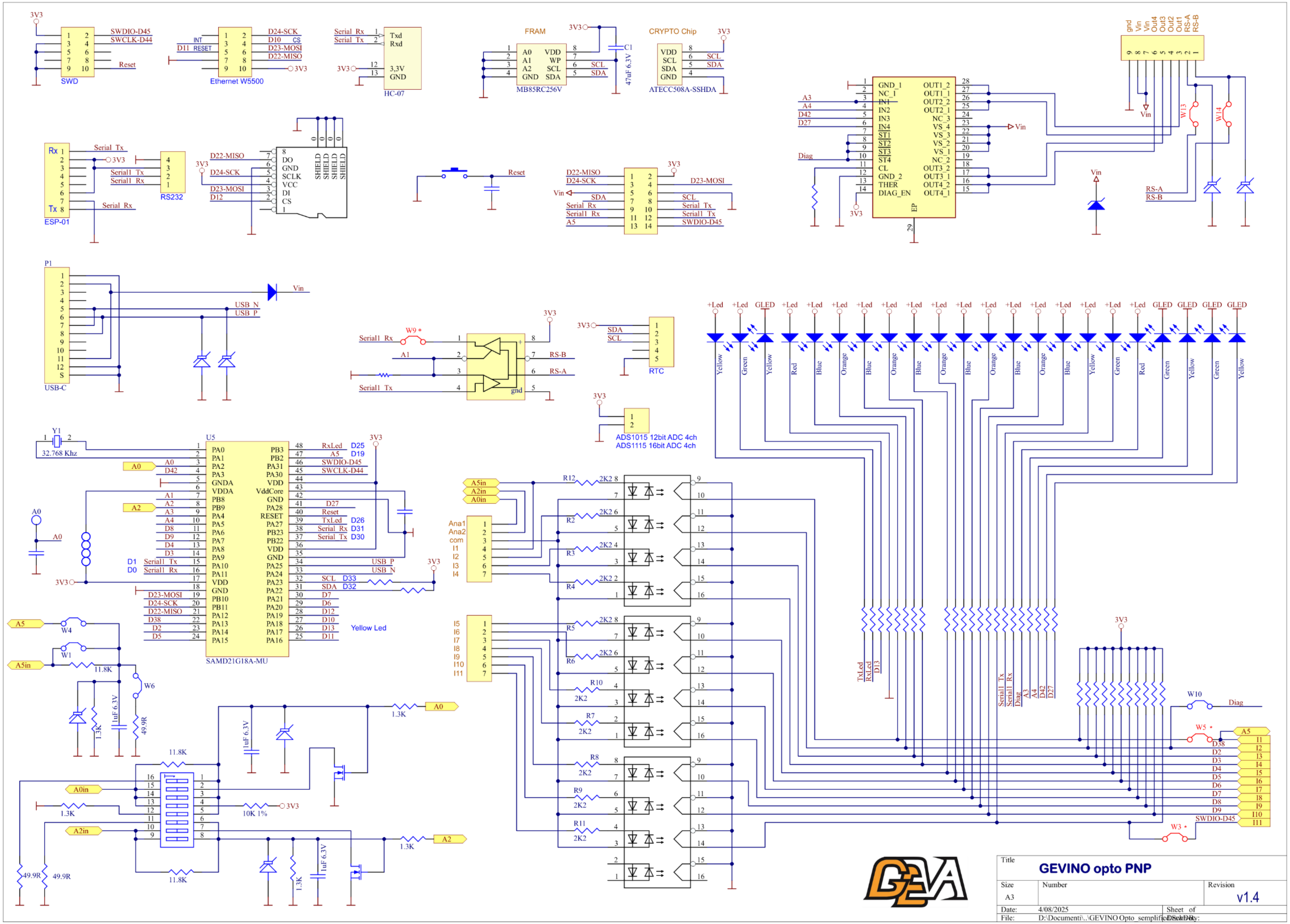
#define dbSerial SerialUSB
#define nexSerial Serial1   (or Serial)

CAN Bus:

MCP2515 module

p10	= VCC	
P7	= GND	
p4	= CS	= D10
P8	= SO	
P6	= SI	
P2	= SCK	
P5	= INT	= D11





SWD
Ethernet W5500
HC-07
FRAM
MB85RC256V
CRYPTO Chip
ATECC508A-SSHDA
ESP-01
RS232
USB-C
RTC
ADS1015 12bit ADC 4ch
ADS1115 16bit ADC 4ch
SAMD21G18A-MU
32.768 Khz
Yellow Led
Title
GEVINO opto PNP
Size
A3
Number
Revision
v1.4
Date: 4/08/2025
Sheet of
File: D:\Documenti\..\GEVINO Opto_semplified.SchDoc
Drawn By: